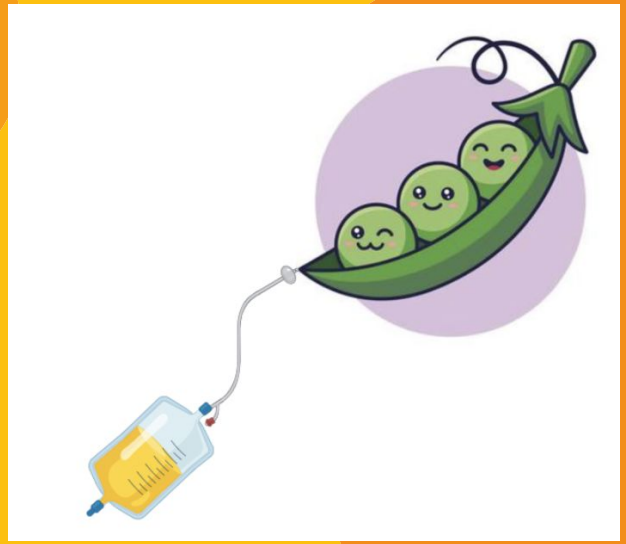
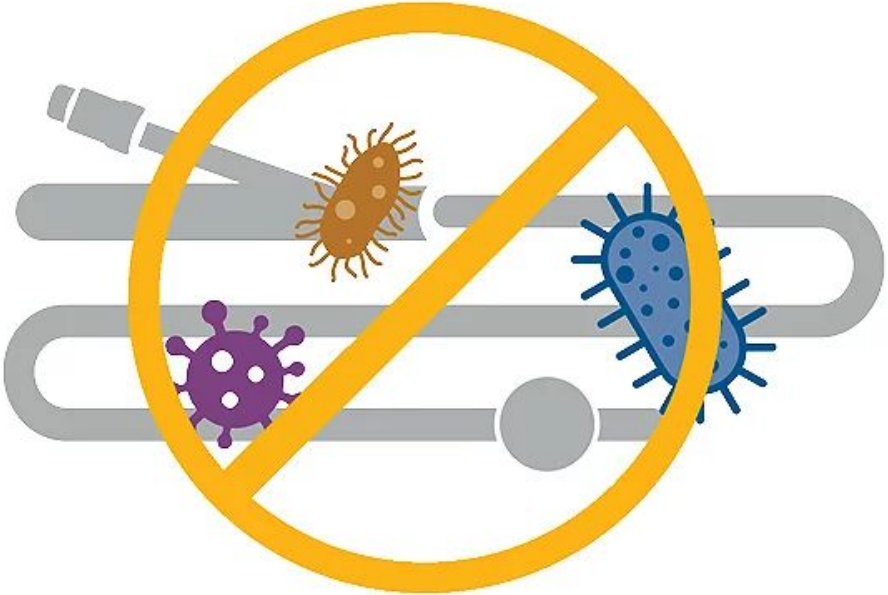
Aptamer-Functionalized Hydrogel Catheter Coating with Embedded Depolymerase to Prevent CAUTI: *PeaS*

Hannah McKee, Ishaan Mehrotra,
Shubhika Munot, Christian Pile, Andy Sun



Overview

Needs Assessment → *Biological Background* → *Primary Design* → *Alternative Designs*

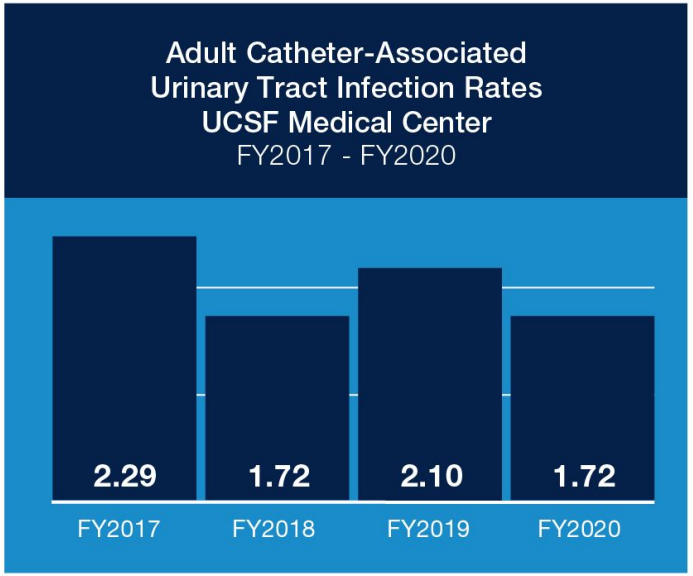


Catheter-associated urinary tract infections (CAUTI) are responsible for **30%** of all healthcare-associated infections[1]!

CAUTI can cause **sepsis, pyelonephritis, urethra stenosis, and bacteremias** [2].

Needs and Markets Assessment

- Patient Need[3] [4] [5]
  - Approximately 450,000 cases annually
  - 75% of hospital-acquired UTIs catheter-related
  - 13,000 deaths annually in the US
- Cost to Patients: $876 to $10,987 per incident
- Market Trends and Treatment of CAUTI
  - Demand for diagnosed at $4.18 billion in 2022
  - Expected to reach $6.32 billion by 2030



Rates are derived from the number of catheter-associated urinary tract infections/1,000 urinary catheter days at among adult patients at UCSF Medical Center.

Current Treatment Options [6][7][8][9]

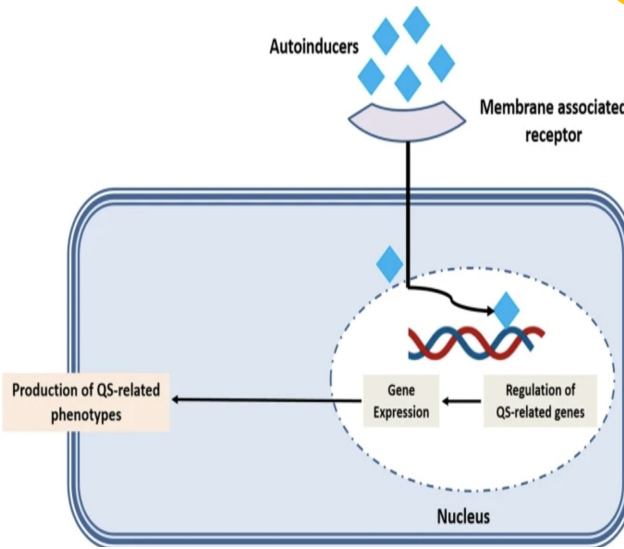
Treatment	Antimicrobial - coated catheters	Hydrogel - coated catheters (existing designs)	Overall Prevention Protocols
Types	Silver-coated, nitrofurazone, minocycline	Dover Foley, Bardex Foley, etc	Sterile insertion, early removal, & bundled care
Cost	$15-30 per unit	$10-15 per unit	N/A
Concern	Short efficacy duration (7-14 days) & rising of resistance	Minimal antimicrobial actions & outdated functionality	Labor-intensive & inconsistent application problems

Needs Statement

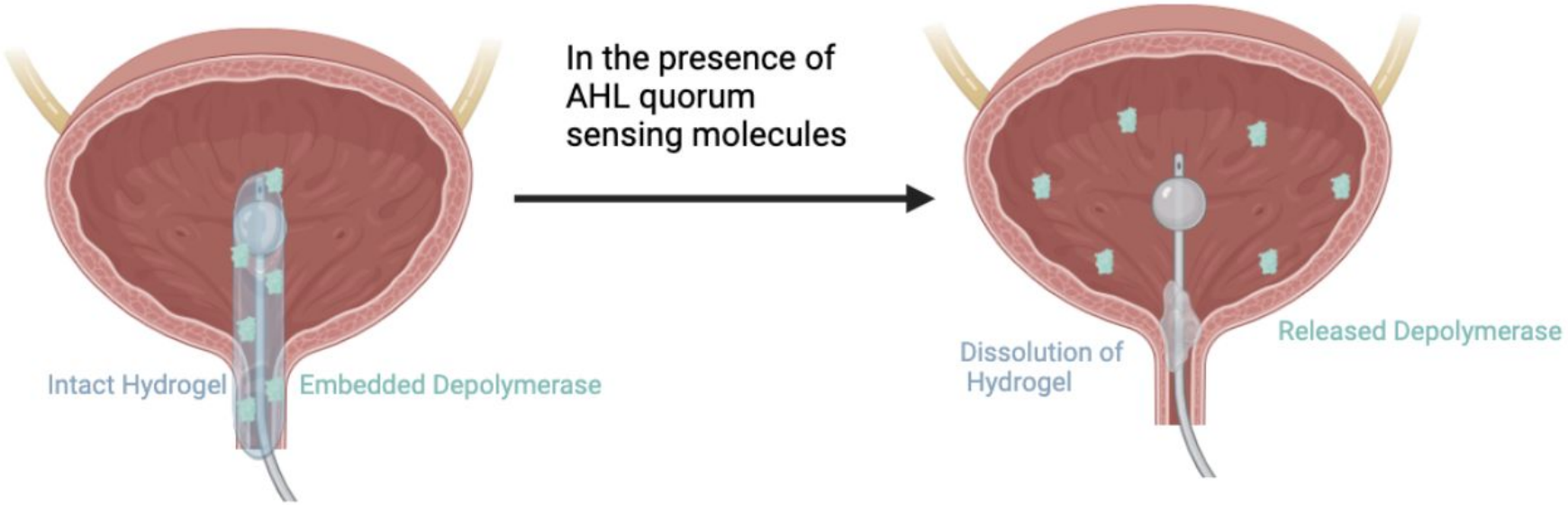
"We aim to develop a CAUTI prevention system via a **dual-faceted** approach with **quorum quenching** and **biofilm depolymerization** incorporated within a **hydrogel coating** for catheters to address **multi-drug resistance of** ***P. mirabilis*** and reduce CAUTI infection rates in **hospital and nursing home populations** .

Relevant Biological Background

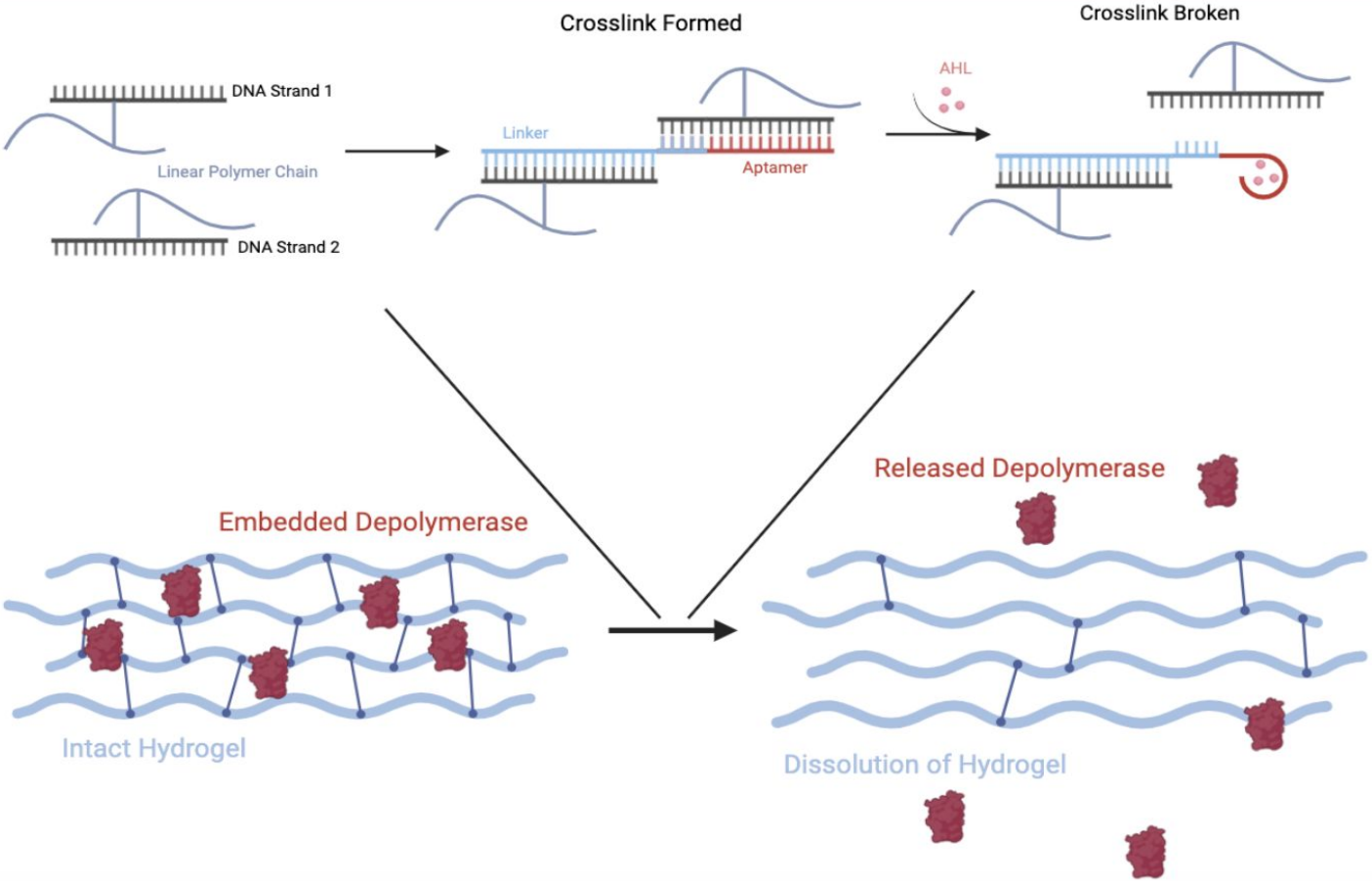
- *P. mirabilis* is the **primary pathogen** responsible for CAUTIs [10]
- **Developed Resistance** to [10]
  - Trimethoprim-sulfamethoxazole, tetracycline, sulfonamides, Beta-Lactams, fluoroquinolones, nitrofurantoin, and more
- Quorum Quenching [11]
  - Interrupts signaling that increases bacterial virulence
  - **AHLs are the quorum sensing molecules that activate LuxI virulence pathways** that enable self replication and swarming
- Phage encoded polymerase protein[12]
  - Proteus **bacteriophage vB_PmiS_Pm-CJR has spike tail protein** that is responsible for biofilm depolymerase
- Aptamers [13]
  - Short single stranded nucleic acid that **binds highly specifically to targets**



Proposed Primary Design (PeaS)

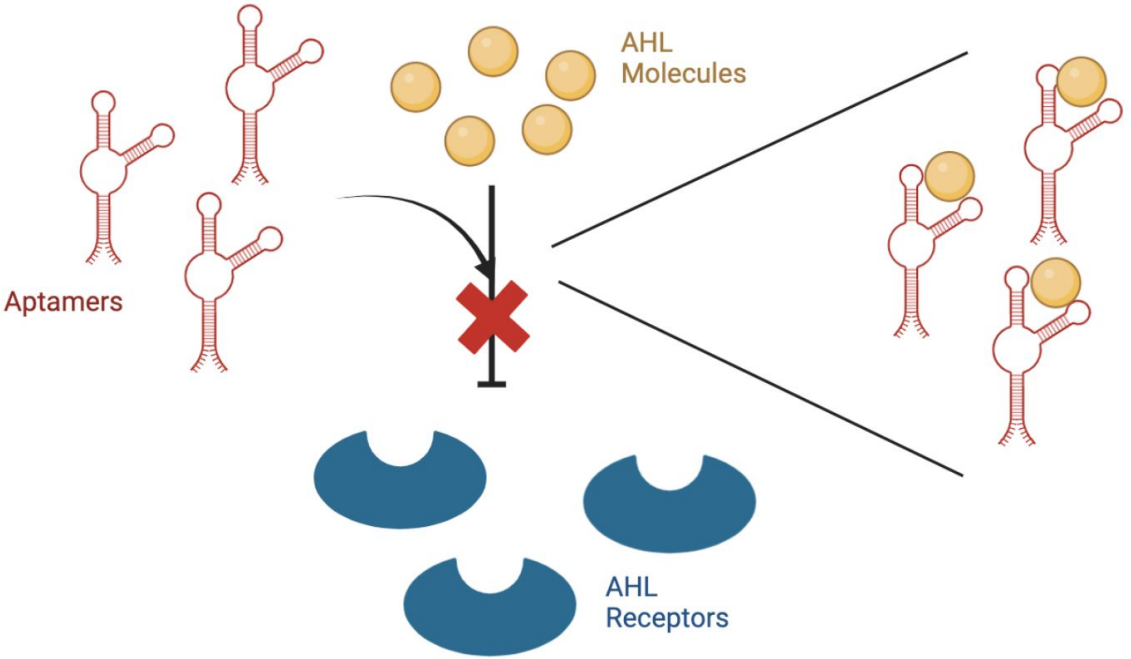


Proposed Primary Design (PeaS)



Alternative Design

- Remove depolymerase to reduce risk to remove risk of competitive binding failures
- Provide more stable and reliable treatment for UTIs



Testing and Implementation

Testing and Implementation

Thank You!

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